DeepTrustxAI Four-Day Cloud and Kubernetes Security Training

Programme Structure

This programme is delivered as two independent tracks:

1. **Cloud Security Foundations, Vulnerabilities and CloudGuard** - 2 days
2. **Kubernetes Security Foundations, Vulnerabilities and CloudGuard** - 2 days

The cloud track can be delivered on its own. The Kubernetes track assumes participants understand basic cloud, Linux, networking, IAM, and containers, but it begins with Kubernetes fundamentals before security testing.

Certification Mapping Basis

CISSP Domains

- D1 Security and Risk Management
- D2 Asset Security
- D3 Security Architecture and Engineering
- D4 Communication and Network Security
- D5 Identity and Access Management
- D6 Security Assessment and Testing
- D7 Security Operations
- D8 Software Development Security

CCSP Domains

- D1 Cloud Concepts, Architecture and Design
- D2 Cloud Data Security
- D3 Cloud Platform and Infrastructure Security
- D4 Cloud Application Security
- D5 Cloud Security Operations
- D6 Legal, Risk and Compliance

The CCSP mapping reflects the ISC2 outline available on 20 June 2026. ISC2 has announced a revised CCSP exam outline effective 1 August 2026; the module mapping should be reviewed against that outline before certification-focused delivery after that date.

Track A: Cloud Security - Two Days

Track Objectives

Participants will be able to:

- Explain cloud service, deployment, responsibility, identity, network, compute, storage, data, and management-plane concepts.
- Recognize how cloud vulnerabilities arise from configuration drift, excessive privilege, public exposure, weak secrets handling, and missing monitoring.
- Model vulnerabilities as connected attack paths rather than isolated findings.
- Run authorized AWS, GCP, IaC, web, secret, and dependency scan workflows.
- Explain how CloudGuard separates the portal, scanner workers, evidence connectors, and GRC reporting.
- Map findings to tenant, asset, evidence, control, owner, remediation, and closure proof.

Cloud Day 1: Foundations and Secure Architecture

Time	Module	Focus	CISSP	CCSP
09:00-09:30	C0 Orientation and lab setup	Authorization, tenants, CloudGuard workflow	D1, D6	D5, D6
09:30-10:45	C1 Cloud computing basics	Service/deployment models, shared responsibility, planes and regions	D1, D3	D1, D3
11:00-12:15	C2 Cloud identity and access	Human/workload identities, federation, roles, keys and least privilege	D5	D1, D3, D5
13:15-14:30	C3 Network, compute and storage	VPC/VNet, segmentation, endpoints, virtual compute and object storage	D2, D3, D4	D2, D3
14:45-15:45	C4 Data security and resilience	Data lifecycle, encryption, secrets, backup, RTO/RPO	D2, D3, D7	D2, D3, D5
15:45-17:00	C5 Secure architecture and IaC	Landing zones, guardrails, logging, IaC and configuration baselines	D1, D3, D7, D8	D1, D3, D4, D5

Cloud Day 2: Vulnerabilities and How CloudGuard Helps

Time	Module	Focus	CISSP	CCSP
09:00-10:15	C6 How cloud vulnerabilities creep in	Misconfiguration lifecycle, drift, shadow assets and ownership gaps	D1, D6, D7, D8	D3, D4, D5, D6
10:30-11:45	C7 Identity, exposure and metadata attack paths	Excess privilege, public access, trust edges, metadata and automation	D3, D4, D5	D1, D3, D4
12:45-13:45	C8 Provider-specific patterns	AWS, GCP, Azure and DigitalOcean control equivalence	D3, D4, D5	D1, D2, D3
13:45-15:00	C9 CloudGuard scanners and job flow	Scanner coverage, credentials, queue, workers and operations	D6, D7	D3, D5
15:15-16:00	C10 Evidence connectors and GRC	Normalization, tenant isolation, control mapping and remediation	D1, D2, D6, D7	D2, D5, D6
16:00-17:00	C11 Cloud capstone	Investigate, scan, ingest, prioritize and brief	D1-D7	D1-D6

Cloud Module Details

C0 Orientation and Lab Setup

Objective: Confirm authorization, assigned tenant, disposable lab environment, and evidence workflow.

Topics:

- Rules of engagement and stop conditions
- CloudGuard portal, scan jobs, workers, connectors, evidence store and GRC
- Tenant-scoped jobs, credentials, assets, findings and reports

Exercise: Trigger a dry-run scan, ingest sample evidence, and verify tenant separation.

C1 Cloud Computing Basics

Objective: Build a common vocabulary before discussing vulnerabilities.

Topics:

- IaaS, PaaS, SaaS and serverless
- Public, private, hybrid and multi-cloud deployment models
- Regions, availability zones, edge locations and fault domains
- Shared responsibility and inherited controls
- Control plane, management plane, data plane and workload plane
- Virtualization, elasticity, automation, APIs and metering

Exercise: Classify twelve responsibilities between customer, provider, and shared ownership.

C2 Cloud Identity and Access

Objective: Understand cloud identity as the primary security boundary.

Topics:

- Human users, groups, roles and federation
- Workload identities, service accounts and managed identities
- Authentication versus authorization
- Temporary credentials versus long-lived keys
- RBAC, ABAC, policy conditions and least privilege
- Privileged access, break-glass and access review

Exercise: Review a permission set, identify broad actions/resources, and produce a least-privilege revision.

C3 Network, Compute and Storage

Objective: Understand the building blocks whose configuration creates or reduces exposure.

Topics:

- Virtual networks, subnets, routes, security groups and network ACLs
- Public/private endpoints, gateways, load balancers and DNS
- Virtual machines, images, containers, functions and metadata services
- Object, block, file, database and ephemeral storage
- Encryption in transit and at rest

Exercise: Draw the intended traffic path for a three-tier cloud application and identify its trust boundaries.

C4 Data Security and Resilience

Objective: Apply data lifecycle, protection, retention and availability concepts to cloud services.

Topics:

- Data classification, location, ownership and lifecycle
- Key, secret and certificate management
- Backup, retention, deletion, legal hold and evidence integrity
- High availability, fault tolerance, RTO and RPO

Exercise: Map a dataset across create, store, use, share, archive and destroy phases with required controls.

C5 Secure Architecture and IaC

Objective: Explain how repeatable architecture reduces recurring vulnerabilities.

Topics:

- Landing zones, account/project structure and guardrails
- Central logging, configuration baselines and asset inventory
- IaC review, policy-as-code, version control and change approval
- Secure defaults, deny guardrails and drift detection

Exercise: Review a Terraform or CloudFormation file and mark preventive, detective and evidence controls.

C6 How Cloud Vulnerabilities Creep In

Objective: Trace vulnerability introduction from design through operation.

Common causes:

- Insecure default or copied template
- Broad temporary permission that becomes permanent
- Manual console change outside IaC
- Public exposure added for troubleshooting
- Long-lived secret copied into code or CI/CD
- New asset not added to inventory or monitoring
- Provider service evolves but the baseline does not
- Ownership and remediation deadlines are unclear

Exercise: Place sample vulnerabilities on the design, build, deploy, operate and retire lifecycle.

C7 Identity, Exposure and Metadata Attack Paths

Objective: Connect individual vulnerabilities into realistic cloud attack paths.

Patterns:

- Public storage to artifact disclosure
- Open service to application weakness
- Application weakness to metadata/runtime identity

- Excess identity permission to sensitive data or control-plane change
- CI/CD token to deployment and cloud role
- Missing logs to delayed detection and weak evidence

Exercise: Build an attack graph and identify the earliest control that breaks the path.

C8 Provider-Specific Patterns

Objective: Recognize equivalent controls across providers.

Exercise: Normalize AWS, GCP, Azure and DigitalOcean mini-cases into a common evidence record.

C9 How CloudGuard Helps: Scanners and Jobs

Capabilities:

- AWS and CloudFox-assisted workflows
- GCP service-account/project scanning
- IaC file and repository scanning
- Web/application checks through Nuclei and OWASP ZAP paths
- Secret and dependency checks through Gitleaks, Trivy and Safety paths
- PostgreSQL-backed `scan_jobs` queue, scheduler, workers, retries, cancellation and dead-letter handling

Exercise: Start a scan job and follow its lifecycle from queued to evidence output.

C10 How CloudGuard Helps: Evidence and GRC

Capabilities:

- Connectors normalize external scanner output
- Evidence includes tenant, source, scanner, job, asset, finding, severity, control and timestamp
- Raw artifact and normalized finding are stored separately
- Compliance dashboard shows framework coverage, evidence count, control status and remediation context
- Multi-tenant design requires `tenant_id` on every scanner-facing record and query

Exercise: Submit a connector payload and map the finding to control, owner, due date and closure proof.

C11 Cloud Capstone

Teams receive an isolated tenant with IaC, IAM, storage, network, logging and CI/CD weaknesses. They must identify the highest-risk connected path, validate it using CloudGuard, ingest evidence, map controls, and deliver a five-minute briefing.

Track B: Kubernetes Security - Separate Two Days

Track Objectives

Participants will be able to:

- Explain container and Kubernetes architecture, objects, identities, networking, storage and scheduling.
- Recognize workload, RBAC, service exposure, network-policy, secret, image and supply-chain vulnerabilities.
- Connect Kubernetes vulnerabilities into cluster attack paths and blast-radius decisions.
- Scan a disposable cluster using kubeconfig and compare live state with IaC.
- Normalize Kubernetes evidence and map it into the CloudGuard compliance dashboard.

Kubernetes Day 1: Foundations and Secure Design

Time	Module	Focus	CISSP	CCSP
09:00-09:30	K0 Orientation and cluster lab	Scope, kubeconfig, namespaces and evidence flow	D1, D6	D5, D6
09:30-10:45	K1 Containers and Kubernetes basics	Images, containers, cluster purpose and orchestration	D3, D8	D1, D3, D4
11:00-12:15	K2 Cluster architecture and objects	Control plane, nodes, pods, controllers and services	D3, D4	D1, D3
13:15-14:30	K3 Identity, RBAC and tenancy	Service accounts, Roles, bindings and namespaces	D5	D3, D5
14:45-15:45	K4 Networking, storage and secrets	CNI, services, policies, volumes, ConfigMaps and Secrets	D2, D3, D4	D2, D3
15:45-17:00	K5 Secure delivery and observability	Images, registries, admission, IaC, logs and audit	D6, D7, D8	D3, D4, D5

Kubernetes Day 2: Vulnerabilities and How CloudGuard Helps

Time	Module	Focus	CISSP	CCSP
09:00-10:15	K6 How Kubernetes vulnerabilities creep in	Weak YAML, drift, defaults, exceptions and ownership	D3, D6, D8	D3, D4, D5
10:30-11:45	K7 Workload and node-risk patterns	Privilege, host access, capabilities and unsafe runtime	D3, D7	D3, D4, D5
12:45-13:45	K8 RBAC, network and secret attack paths	Broad bindings, exposed services, missing policies and secret access	D2, D4, D5	D2, D3, D5
13:45-15:00	K9 Supply-chain and IaC drift	Images, dependencies, manifests, admission and live-state drift	D6, D8	D3, D4, D5
15:15-16:00	K10 CloudGuard Kubernetes scanner and GRC	Kubeconfig, live inventory, RBAC, network and evidence mapping	D6, D7	D3, D5, D6
16:00-17:00	K11 Kubernetes capstone	Scan, prioritize, remediate and prove closure	D1-D8	D1-D6

Kubernetes Module Details

K0 Orientation and Cluster Lab

Confirm assigned kubeconfig/context, namespace, disposable cluster boundaries, scanner endpoint, evidence tenant and cleanup process.

K1 Containers and Kubernetes Basics

Topics: image, container, runtime, registry, orchestration, desired state, declarative configuration and reconciliation.

Exercise: Trace an application from image build to registry, pod, service and external request.

K2 Cluster Architecture and Objects

Topics: API server, etcd, scheduler, controller manager, kubelet, container runtime, Pods, Deployments, StatefulSets, DaemonSets, Jobs, CronJobs and Services.

Exercise: Match each object to its control-plane component, lifecycle and failure impact.

K3 Identity, RBAC and Tenancy

Topics: users, groups, service accounts, tokens, Roles, ClusterRoles, RoleBindings, ClusterRoleBindings, namespace isolation and external identity integration.

Exercise: Calculate effective access for a prepared service account and reduce its permissions.

K4 Networking, Storage and Secrets

Topics: cluster networking, CNI, DNS, ClusterIP, NodePort, LoadBalancer, ingress, NetworkPolicy, persistent volumes, storage classes, ConfigMaps and Secrets.

Exercise: Draw allowed traffic, identify missing policy boundaries, and classify mounted data.

K5 Secure Delivery and Observability

Topics: image provenance, registry controls, vulnerability scanning, signing, admission control, policy-as-code, IaC, audit logs, events and runtime monitoring.

Exercise: Review a deployment pipeline and identify preventive, admission, runtime and evidence controls.

K6 How Kubernetes Vulnerabilities Creep In

Common causes:

- Copied YAML with privileged or root settings
- Temporary exception to admission policy
- Broad service account or cluster binding
- Service changed to NodePort/LoadBalancer for troubleshooting
- Secret mounted into an unnecessary workload
- NetworkPolicy omitted for a new namespace
- Image tag floats to an unreviewed version
- Manual cluster change creates drift from Git/IaC
- Scanner warning has no owner or remediation deadline

K7 Workload and Node-Risk Patterns

Patterns: privileged containers, hostPID/hostNetwork, hostPath, dangerous capabilities, root execution, writable root filesystem, missing limits and node-level access.

Exercise: Review prepared manifests and prioritize findings by escape potential and node blast radius.

K8 RBAC, Network and Secret Attack Paths

Patterns: broad ClusterRoleBindings, token auto-mount, exposed services, missing east-west policy, namespace crossover and secret visibility.

Exercise: Build a path from exposed service to workload identity to secret or cluster action.

K9 Supply-Chain and IaC Drift

Patterns: vulnerable base image, leaked registry credential, unsigned image, mutable tag, hardcoded secret, unsafe manifest and live-state drift.

Exercise: Compare repository manifests with live resources and create an evidence-backed drift finding.

K10 How CloudGuard Helps Kubernetes

Capabilities:

- Validate kubeconfig and connect to the assigned live cluster
- Query pods, services, namespaces, deployments, jobs and cronjobs
- Review RoleBindings and ClusterRoleBindings
- Identify exposed services and missing NetworkPolicies
- Detect risky workload settings and secret exposure patterns
- Compare live resources with Kubernetes IaC manifests
- Enrich managed-cluster context when AWS/GCP credentials exist
- Store findings as tenant-scoped evidence and map them to controls

Exercise: Run the scanner, inspect job/evidence lineage, prioritize findings and verify one remediation through rescan.

K11 Kubernetes Capstone

Teams receive a disposable cluster with intentional workload, RBAC, service, network-policy, secret, image and drift weaknesses. They must identify the highest-risk path, validate it through CloudGuard, map the evidence to controls and owners, remediate selected findings, and prove closure.

Shared Assessment Model

Area	Points
Foundation and architecture understanding	15
Accurate vulnerability/root-cause analysis	20

Area	Points
Attack-path and blast-radius reasoning	20
Scanner validation and evidence quality	20
CISSP/CCSP control mapping	10
Remediation and closure proof	10
Executive communication	5

Official References

- ISC2 CISSP Exam Outline, effective 15 April 2024: `https://www.isc2.org/certifications/cissp`
- ISC2 CCSP Exam Outline and announced 1 August 2026 update: `https://www.isc2.org/certific`
- DeepTrustxAI official logo: `https://deeptrustxai.com/images/darkmode.png`